

Portfolio Construction and Out-of-Sample Tests

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1 Data and Methodology

This section outlines the data foundation and empirical methodology employed in our analysis. We begin by describing the construction of the dataset, including the sample selection, filtering procedures, and the computation of returns and covariance matrices. Subsequently, we introduce a systematic framework to determine the effective number of latent factors that drive asset returns, leveraging statistical tests for skewness to identify meaningful components. Building upon this factor structure, we then detail several approaches for estimating return covariance matrices, ranging from shrinkage-based estimators to factor-model representations and eigenvector-regularized formulations. Together, these elements form the methodological backbone for our empirical results and ensure the robustness of subsequent portfolio and risk analyses.

1.1 Data Source

Our data consists of the largest 500 US stocks by market capitalization, with a time period from 2015 to 2024. We use the CRSP Database on WRDS to obtain stock prices and market capitalization data. Securities without a complete history are dropped, and the remaining stocks are used to construct the return covariance matrix. Stock list and capitalization are

updated monthly, and we use a look-back window of three months, which provides a number of observations $n \approx 63$ for each period.

1.2 A Systematic Approach to Identify the Number of Factors

Given a dataset $X \in \mathbb{R}^{n \times d}$ where $d \gg n$, we apply the following procedure:

1. Perform PCA on X .
2. Compute the principal component scores and sequentially remove the first k components.
3. Compute the squared residual lengths:

$$R_j(k) = \frac{1}{d} \left\| X_j - \sum_{i=1}^k u_i u_i^T X_j \right\|^2 \quad (1)$$

4. Conduct the triples test for skewness to detect left- or right-skewness in residual squared lengths.
5. Determine the number of meaningful factors as the smallest k where the test fails to reject the null hypothesis at a significance level α .

1.2.1 Triples Test for Skewness

Test Statistic Given a sample of n independent observations $y_1, y_2, \dots, y_n$, we define the test statistic based on the following kernel function:

$$f^*(y_i, y_j, y_k) = \text{sign}(y_i + y_j - 2y_k) + \text{sign}(y_i + y_k - 2y_j) + \text{sign}(y_j + y_k - 2y_i), \quad (2)$$

For a given dataset, the test statistic is computed as:

$$T_n = \frac{1}{\binom{n}{3}} \sum_{i < j < k} f^*(y_i, y_j, y_k). \quad (3)$$

Standardization Under the null hypothesis of symmetry, T_n is asymptotically normal with mean 0 and variance:

$$\sigma_{\hat{T}}^2 = \frac{2(n-1)}{9n}. \quad (4)$$

Thus, the standardized test statistic is:

$$Z_n = \frac{T_n}{\sigma_{\hat{T}}} = \frac{T_n}{\sqrt{\frac{2(n-1)}{9n}}}. \quad (5)$$

P-value Computation The test statistic Z_n follows an asymptotic standard normal distribution, so the two-sided p-value is given by:

$$p = 2(1 - \Phi(|Z_n|)), \quad (6)$$

where $\Phi(x)$ is the cumulative distribution function of the standard normal distribution. We use this p-value to determine whether the data exhibit significant skewness. If $p > \alpha$ (e.g. 0.5), we determine left-skewness, and vice-versa.

1.2.2 D'Agostino's Skewness Test

Let the residuals be $R_1(k), \dots, R_n(k)$ with sample mean $\bar{R} = \frac{1}{n} \sum_{j=1}^n R_j(k)$. Define the second and third central moments $m_2 = \frac{1}{n} \sum_{j=1}^n (R_j(k) - \bar{R})^2$ and $m_3 = \frac{1}{n} \sum_{j=1}^n (R_j(k) - \bar{R})^3$. The sample skewness is $b_1 = m_3/m_2^{3/2}$.

Apply the Johnson S_U transformation (valid for $n \geq 8$) to obtain an approximately standard normal statistic:

$$Y = b_1 \sqrt{\frac{(n+1)(n+3)}{6(n-2)}}, \quad \beta_2 = \frac{3(n^2 + 27n - 70)(n+1)(n+3)}{(n-2)(n+5)(n+7)(n+9)},$$

$$W = \sqrt{2(\beta_2 - 1)}, \quad \alpha = \sqrt{\frac{2}{W^2 - 1}}, \quad \delta = \frac{1}{\sqrt{\log W}},$$

$$Z = \delta \ln \left(\frac{Y}{\alpha} + \sqrt{1 + \left(\frac{Y}{\alpha} \right)^2} \right).$$

Under the null of normality, Z is approximately $\mathcal{N}(0, 1)$. The one-sided p -value for right-skewness is

$$p_k^{(D)} = 1 - \Phi(Z),$$

where $\Phi(\cdot)$ is the standard normal CDF. Small $p_k^{(D)}$ rejects normality in favor of positive skew in $\{R_j(k)\}$.

1.2.3 Results from the Factor-Counting Procedure

The results obtained for each month using a three-month rolling window are presented in Figure 1. Given that the two skewness tests yield identical outcomes, we adopt the more refined alternative, namely, the triples test. We further report two distinct implementations of this procedure: one based on the correlation matrix and the other on the covariance matrix. These are represented by the orange and blue curves, respectively, in Figure 1.

1.3 Covariance Matrices Construction

In this subsection, we present four different methods we use to construct return covariance matrices. The first method is to compute the covariance matrix directly from the return matrix. While the other three will be discussed in the following subsections.

1.3.1 Data Preparation

Let $X_{\text{raw}} \in \mathbb{R}^{p \times n}$ be the monthly return panel. The pipeline:

1. **Row filtering:** drop any asset row containing at least one NaN.

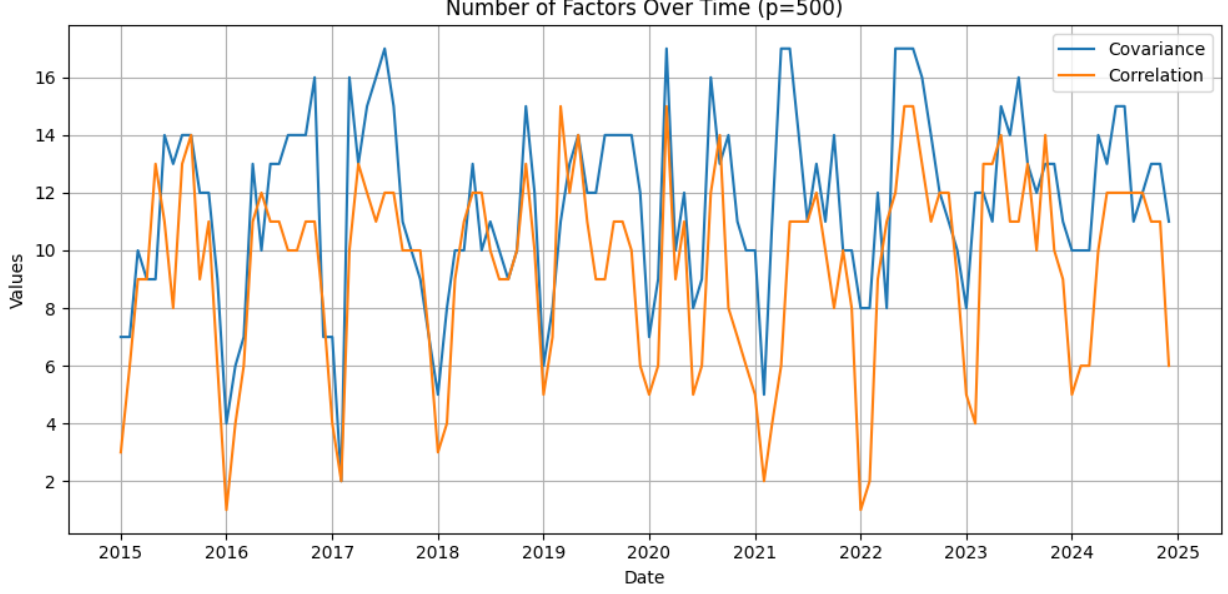


Figure 1

2. **Demeaning by asset:** with $\bar{x}_i$ the mean of row i ,

$$X = X_{\text{raw}} - [\bar{x}_1 \mathbf{1}_n; \dots; \bar{x}_p \mathbf{1}_n].$$

The sample covariance uses $S = \frac{1}{n} X X^\top$.

Effective factor count. If a month has k factors indicated by the method in Section 1.2, we enforce

$$k_{\text{eff}} = \max(0, \min(k, p - 1, n - 1)),$$

avoiding rank issues in very short samples.

1.3.2 Estimator 1: Ledoit–Wolf Shrinkage

The Ledoit–Wolf estimator shrinks S to a scaled identity:

$$\Sigma_{\text{LW}} = (1 - \delta)S + \delta F, \quad F = \mu I_p, \quad \mu = \frac{\text{tr}(S)}{p}.$$

The shrinkage intensity

$$\delta = \text{clip}\left(\frac{\hat{\phi}}{\hat{\gamma}}, 0, 1\right),$$

with

$$\hat{\phi} = \frac{1}{n} \sum_{t=1}^n \|x_t x_t^\top - S\|_F^2, \quad \hat{\gamma} = \|S - F\|_F^2,$$

is computed directly from demeaned returns $\{x_t\}$. If $\hat{\gamma} \leq 0$, the code sets $\delta = 1$ for stability.

The final matrix is re-symmetrized.

1.3.3 Estimator 2: PCA Factor Covariance

Compute the top- k_{eff} eigenpairs of S :

$$S \approx U_k \Lambda_k U_k^\top + \Psi, \quad \Psi = \text{diag}\left(\text{diag}(S - U_k \Lambda_k U_k^\top)\right),$$

with an ε -floor ($\varepsilon = 10^{-12}$ by default) applied element-wise so that idiosyncratic variances remain strictly positive. This yields a positive semidefinite factor+diagonal structure while preserving S 's diagonal in aggregate.

1.3.4 Estimator 3: James–Stein Eigenvector Shrinkage + Factor Covariance

This variant modifies the *eigenvectors* before reconstructing a factor covariance:

- (a) Compute top- k_{eff} eigenpairs (U_k, Λ_k) of S .
- (b) For each sample eigenvector u_j with eigenvalue λ_j , let $m(u_j) = \frac{1}{p} \mathbf{1}_p^\top u_j$ be the average entry of u_j and write $\mathbf{1}_p$ for the all-ones vector. Apply James–Stein shrinkage (per the screenshot) to obtain

$$u_j^{\text{JSE}} = m(u_j) \mathbf{1}_p + c_j^{\text{JSE}} (u_j - m(u_j) \mathbf{1}_p), \quad c_j^{\text{JSE}} = 1 - \frac{\nu^2}{s^2(u_j)},$$

where

$$s^2(u_j) = \frac{\lambda_j^2}{p} \sum_{i=1}^p (u_{ij} - m(u_j))^2, \quad \nu^2 = \frac{\text{tr}(S) - \lambda_j^2}{p(n-1)}.$$

- (c) Re-orthogonalize the shrunken columns via QR to obtain $\tilde{U}_k$, then align each column's sign with the original u_j (to avoid arbitrary flips).
- (d) Form the factor covariance

$$\Sigma_{\text{JSE}} = \tilde{U}_k \Lambda_k \tilde{U}_k^\top + \text{diag}\left(\text{diag}(S - \tilde{U}_k \Lambda_k \tilde{U}_k^\top)\right),$$

again applying the ε floor to idiosyncratic variances.

Intuitively, when the dispersion $s^2(u_j)$ dominates ν^2 , $c_j^{\text{JSE}} \approx 1$ and little shrinkage occurs; when $s^2(u_j)$ is small relative to ν^2 , the vector is pulled toward its mean direction $m(u_j)\mathbf{1}_p$.

1.4 Portfolio Construction

This section specifies a reproducible procedure to turn every month's full-rank covariance matrix Σ_t into several investable portfolios. It applies identically to all four construction methods. Begin with the simplest portfolio (global minimum variance), then add structure (long-only box constraints), and finally incorporate return forecasts and alternative risk objectives.

Data and alignment. On analysis date t , read $\Sigma_t \in \mathbb{R}^{p \times p}$ and its ordered asset list $\mathcal{I}_t = \{1, \dots, p\}$. Ensure the ordering of rows and columns is identical and that assets with missing data were already removed upstream. If Σ_t is near-singular, add a small ridge εI (e.g., $\varepsilon = 10^{-6}$) to stabilize inverses. Define the all-ones vector $\mathbf{1} \in \mathbb{R}^p$.

Common constraints and knobs. Unless otherwise stated, we solve for weights $w \in \mathbb{R}^p$ using:

$$\mathbf{1}^\top w = 1, \quad \ell \leq w \leq u,$$

where (ℓ, u) are lower/upper bounds. Defaults: (i) Long-only: $\ell = \mathbf{0}$, $u = 0.05$ (5% cap) or no cap; (ii) Long-short with leverage budget: replace box with $\|w\|_1 \leq L$ (e.g., $L = 1.5$)

and allow $\ell = -u$. Optionally add a small ℓ_2 regularizer $\lambda\|w\|_2^2$ (e.g., $\lambda = 10^{-4}$) to spread weights and improve stability.

Portfolio A (Baseline): Global Minimum Variance (GMV)

Unconstrained closed form. If shorting is allowed and no bounds bind,

$$w_t^{\text{GMV}} = \frac{\Sigma_t^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma_t^{-1}\mathbf{1}}.$$

Constrained (long-only / box). Solve the quadratic program (QP)

$$\min_w w^\top \Sigma_t w + \lambda \|w\|_2^2 \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad \ell \leq w \leq u.$$

Implement with any convex QP solver. Save weights using the exact asset order $\mathcal{I}_t$.

Portfolio B: Target-Volatility MV with Zero-Mean Prior

When expected returns are unreliable, a practical step-up from GMV is to anchor on a zero-mean prior and choose risk via γ :

$$\min_w \frac{\gamma}{2} w^\top \Sigma_t w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad \ell \leq w \leq u.$$

Here $\gamma > 0$ tunes aggressiveness; higher γ yields lower ex-ante variance. Choose γ to target a pre-set ex-ante volatility level σ^* by line search on γ so that $\sqrt{w^\top \Sigma_t w} \approx \sigma^*$ (e.g., 10% annualized).

1.5 Portfolio Tests

To evaluate the performance of different covariance matrix estimation methods, we conduct out-of-sample portfolio tests using monthly realized returns. On each analysis date, several covariance matrices are estimated through alternative construction methods. These matrices

are then used to optimize portfolios under specific objectives such as global minimum variance (GMV) or target variance minimum variance (TVMV).

After constructing each optimized portfolio at month-end, its performance is tested during the subsequent month. Specifically, portfolio weights derived from the optimization step are applied to the realized asset returns of the next month to compute the portfolio's realized return. This procedure yields a time series of out-of-sample returns for each portfolio, allowing comparison of different covariance estimation methods.

The primary evaluation metrics are the mean and variance of the realized portfolio returns. Lower realized variance and higher mean return indicate better performance of the underlying covariance estimation technique. This systematic testing framework provides an empirical basis for assessing which covariance construction methods lead to more stable and efficient portfolio outcomes.

2 Simulated Data

2.1 Single-Factor Model

Daily returns are generated via a standard single-factor specification:

$$r_{i,t} = \alpha_i + \beta_i F_t + \varepsilon_{i,t}, \quad i = 1, \dots, 500, \quad t = 1, \dots, 63,$$

where:

- F_t is a common factor shock for day t .
- α_i is an asset-specific drift term (daily alpha).
- β_i is the asset's loading on the common factor.
- $\varepsilon_{i,t}$ is an idiosyncratic shock, independent across assets and time and independent of F_t .

2.2 Parameterization

The cross-sectional parameters $\{\alpha_i, \beta_i, \sigma_i^\varepsilon\}$ are drawn once and then held fixed for the entire experiment:

$$\alpha_i \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2), \quad \beta_i \sim \mathcal{N}(\mu_\beta, \sigma_\beta^2), \quad \sigma_i^\varepsilon \sim \max\{\mathcal{N}(\mu_\sigma, \sigma_\sigma^2), \underline{\sigma}\}.$$

We use the following values:

$$\mu_F = 0.0003, \quad \sigma_F = 0.01, \quad \mu_\alpha = 0.00005, \quad \sigma_\alpha = 0.0002, \quad \mu_\beta = 1.0, \quad \sigma_\beta = 0.3,$$

$$\mu_\sigma = 0.012, \quad \sigma_\sigma = 0.004, \quad \underline{\sigma} = 0.003.$$

The common factor process is i.i.d. Gaussian:

$$F_t \sim \mathcal{N}(\mu_F, \sigma_F^2),$$

drawn independently for each day within each month. Idiosyncratic shocks are Gaussian and independent across assets and time. In the current implementation, all assets share a common idiosyncratic standard deviation equal to the mean value $\mu_\sigma = 0.012$:

$$\varepsilon_{i,t} \sim \mathcal{N}(0, \mu_\sigma^2).$$

All random draws use a fixed seed for reproducibility.